Share A domain

D : banks s0_*, BFU0 + BFU1

Share B domain

D ¹ : banks s1_*, BFU2 + BFU3

zeta_addr

Twiddle ROM
128 × 16 bit (public, shared)
crosses boundary safely (no secret)

LFSR-16
dummy_a/b/w (L1)
rand_seed → shuffle (L2)
public entropy

rand_seed (shuffle_mask)

ext_w

ext_r

ext_share_sel

zeta (w)

zeta (w)

s0_even
128 × 16 bit
addr[0] = 0
share A

a, b

a_out, b_out

BFU0
5-stage Mont.
share A, even
L1 mux on inputs

BFU2
5-stage Mont.
share B, even
L1 mux on inputs

a, b

a_out, b_out

s1_even
128 × 16 bit
addr[0] = 0
share B

s0_odd
128 × 16 bit
addr[0] = 1
share A

a, b

a_out, b_out

BFU1
5-stage Mont.
share A, odd
L1 mux on inputs

BFU3
5-stage Mont.
share B, odd
L1 mux on inputs

a, b

a_out, b_out

s1_odd
128 × 16 bit
addr[0] = 1
share B

dummy_a/b/w

dummy_a/b/w

bf_active

bf_active

DOMAIN BOUNDARY
no combinational path crosses during NTT computation

Controller (FSM)
j += 2 per iteration · L2 shuffle: gs = gc mask
IDLE → READ → WAIT → WRITE → NEXT
shared addr/we for both shares (no secret)

start

done

addr / we (share A)

addr / we (share B)

data path · zeta (twiddle) · dummy operands · control path · domain boundary